Depth-Scaled Gaussian Initialization for Trainable Quantum Neural Networks

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***Abstract*—Training deep quantum neural networks (QNNs) is challenging because barren plateaus cause gradients to diminish exponentially as the number of qubits and circuit depth increase. Zhang et al. (NeurIPS 2022) demonstrated that a Gaussian initialization with a variance that decreases with depth can maintain polynomially large gradients. However, they tested only physics and chemistry tasks. In this project, we reproduce their results and evaluate their initialization strategy on a QNN classifier in both noiseless and noisy hardware simulations. In preliminary tests on a 12-qubit classifier of 8×8 handwritten digits, Gaussian initialization gives a squared gradient norm 147 times larger than uniform initialization.**

***Index Terms*—Quantum machine learning, quantum neural networks, parameter initialization, vanishing gradients, barren plateaus, noisy simulation.**

I. Problem

Weight initialization determines whether a deep model can be trained, which is why classical networks use schemes such as Xavier and He initialization. Quantum neural networks face a more severe form of this problem, known as the barren plateau. With uniformly random initial parameters, the gradient variance shrinks exponentially with the number of qubits [1]. QNN classifiers inherit this behavior from deep circuits and from the way the input data are encoded [2]. On quantum hardware, gradients are estimated from repeated measurements, so vanishing gradients are also expensive to measure. For example, a gradient variance of 2^{-15} at 15 qubits means a typical gradient of about 0.0055, which takes roughly 33,000 measurements to resolve. Zhang et al. [3] remedy this by drawing the initial parameters from a Gaussian whose variance shrinks with depth, and prove that gradients then decay only polynomially. However, they test this rule only on physics and chemistry tasks, with noise added only to the gradients. This project tests it on deep QNN classifiers and amid simulated hardware noise.

II. Quantum Computing Background

The following terms are used throughout this proposal.

- *Qubit:* A quantum bit analogous to a classical bit, represented as a vector in $\mathbb{C}^2$ with complex amplitudes [4].
- *Quantum gate:* An operation on one or more qubits. The single-qubit rotations $R_X(\theta) = e^{-i\theta X}$ and $R_Y(\theta) = e^{-i\theta Y}$, with Pauli matrices X and Y, are set by an angle θ. The two qubits are entangled by a two-qubit CZ gate [4].
- *Observable:* A measurable quantity, such as the Pauli-Z matrix. Its expectation value is estimated by averaging repeated measurements, called shots [4].
- *Quantum neural network (QNN):* A circuit of data-encoding gates followed by layers of rotations and CZ gates, whose output is the expectation value of an observable [5]. Its rotation angles are the trainable weights.
- *Parameter-shift rule:* A method that computes each partial derivative of an expectation value from two circuit evaluations at shifted angles [6].
- *Hamiltonian:* An observable whose expectation value is the energy of a physical system. Its smallest eigenvalue is the ground energy [4]. Its locality S is the largest number of qubits any of its terms acts on.
- *Device noise:* Errors on real hardware, namely imperfect gates, decoherence, and readout errors [4]. Qiskit simulates them with noise models calibrated from IBM quantum processors [7].

III. Method Under Study

This project studies the initialization rule of Zhang et al. [3], summarized below.

- *Initialization rule:* Every trainable angle is drawn from $\mathcal{N}(0, \gamma^2)$ with $\gamma^2 = 1/(4S(L+2))$, where L is the number of rotation layers before the final two.
- *Guarantee:* Theorem 4.1 in [3] bounds the expected squared gradient norm from below by $L/(S^S(L+2)^{S+1})$, times a factor set by the input state. The bound decays polynomially with depth, rather than exponentially with the number of qubits.
- *Original experiment:* The ground energy estimation of a 15-qubit Heisenberg chain ($S = 2$) using a 300-parameter circuit ($L = 18$, $\gamma^2 = 1/160$), against uniform $[0, 2\pi]$ and zero initialization.

IV. Plan

The project consists of three studies, in priority order.

- *S1, reproduction:* the Heisenberg experiment of [3] in PennyLane, with gradient descent and Adam, exact and noisy gradients, and 5 seeds per setting.
- *S2, QNN classifier:* a QNN that classifies handwritten digits, with Gaussian, uniform, and zero initialization compared as the number of qubits grows.
- *S3, noisy hardware simulation:* the same comparisons in Qiskit with IBM device noise models at 5 and 7 qubits.

V. Contribution

To our knowledge, this is the first test of the rule of [3] on a deep QNN classifier, where barren plateaus occur, and under calibrated device noise. A recent benchmark of initializations

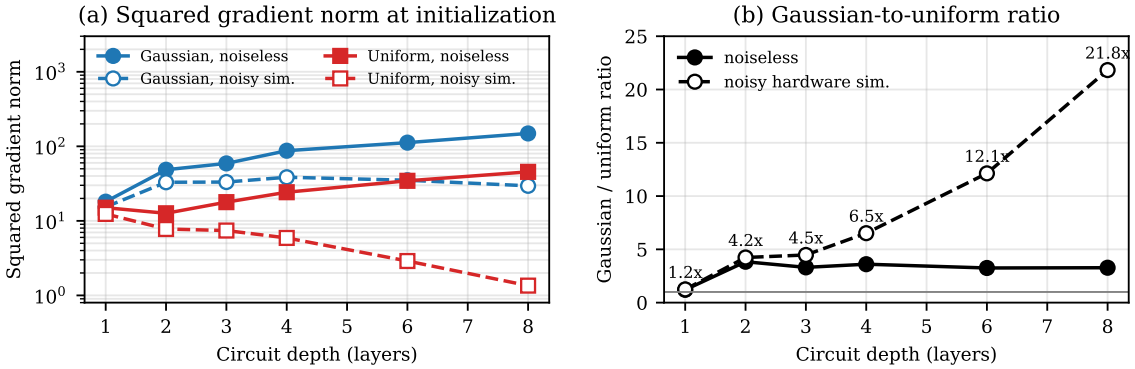


Fig. 1. Gradient at initialization for a 5-qubit Heisenberg chain, with and without device noise. (a) Squared gradient norm. (b) Ratio of the Gaussian to the uniform squared gradient norm.

for QNN image classifiers [8] used two-layer circuits, found no severe barren plateaus, and modeled no hardware noise. Device noise causes barren plateaus of its own [9]. The closest study of initialization under noise uses idealized depolarizing noise and a different small-angle start [10].

TABLE I
SQUARED GRADIENT NORM OF THE QNN CLASSIFIER AT INITIALIZATION.

Qubits	Params	Gaussian	Uniform	Zero	G/U
4	80	5.68	3.98	1.03	1.4
6	120	10.19	1.35	1.06	7.5
8	160	1.27	0.45	1.00	2.8
10	200	6.68	0.12	1.07	55
12	240	8.93	0.061	1.17	147

VI. Preliminary Work

To check for a barren plateau, we built a 10-layer QNN that classifies the handwritten digits 3 and 8, and measured its squared gradient norm at initialization, averaged over 5 random initializations and 16 training images (Table I). With uniform initialization, the squared gradient norm falls 65 times from 4 to 12 qubits, while Gaussian initialization keeps it between 1.3 and 10.2. As a result, the Gaussian-to-uniform ratio (G/U) reaches 147 at 12 qubits. In a short 8-qubit training run, all three initializations reach about 0.90 test accuracy, but Gaussian initialization has the lowest test error (0.27 against 0.46 for uniform). A reproduction check with gradient descent reaches energies of -19.2 after 10 iterations and -22.8 after 100, against about -19 and -22.8 in Fig. 2(a) of [3]. To test the effect of device noise, we computed the energy gradient of a 5-qubit Heisenberg chain at initialization, with and without an IBM processor noise model, averaged over 8 random initializations (Fig. 1). With noise, the squared gradient norm under uniform initialization drops 9-fold from 1 to 8 layers, while under Gaussian initialization it stays between 15 and 39. As a result, the Gaussian-to-uniform ratio stays near 3.3 without noise but grows to 21.8 with noise.

VII. Experiment Plan

Hypothesis. On a 12-qubit QNN classifier, Gaussian initialization reaches a target test accuracy in fewer training steps than uniform initialization, and keeps this advantage in the noisy hardware simulation at smaller sizes.

Proxy. The statistical metrics are test accuracy, training loss per step, and steps to reach the target accuracy. The hardware metrics are wall-clock time per step and circuit evaluations per gradient, which are $2 \times 240 = 480$ per image at 12 qubits.

Protocol. Each initialization is trained with Adam (learning rate 0.01, batch 16, 100 steps) on a fixed train/test split over 5 seeds.

Expected results. Gaussian initialization should train faster than uniform initialization at 10 or more qubits, and the gap should persist or widen under noise. Negative results are acceptable.

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